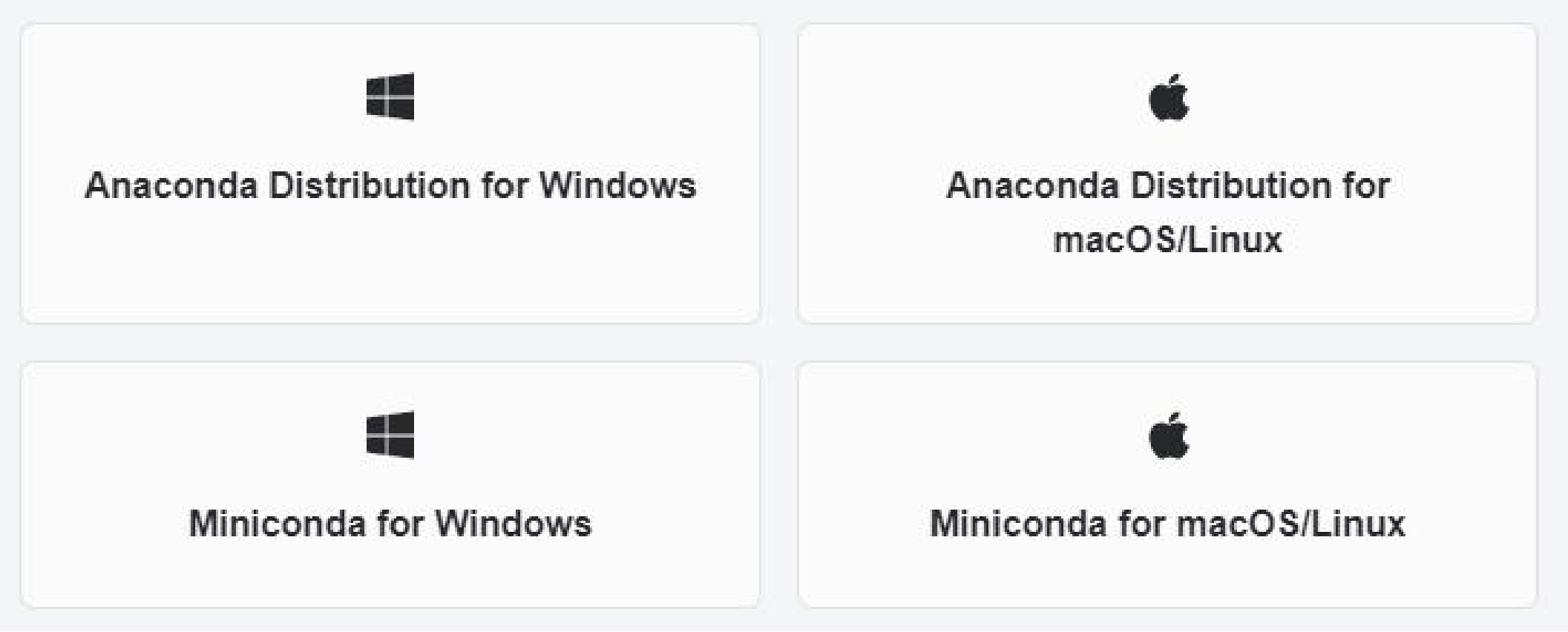
CS5187: Vision & Image

香港城市大學(東莞)
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Zhu Zhiyu

Tutorial 01: Python environment set up and basic image processing

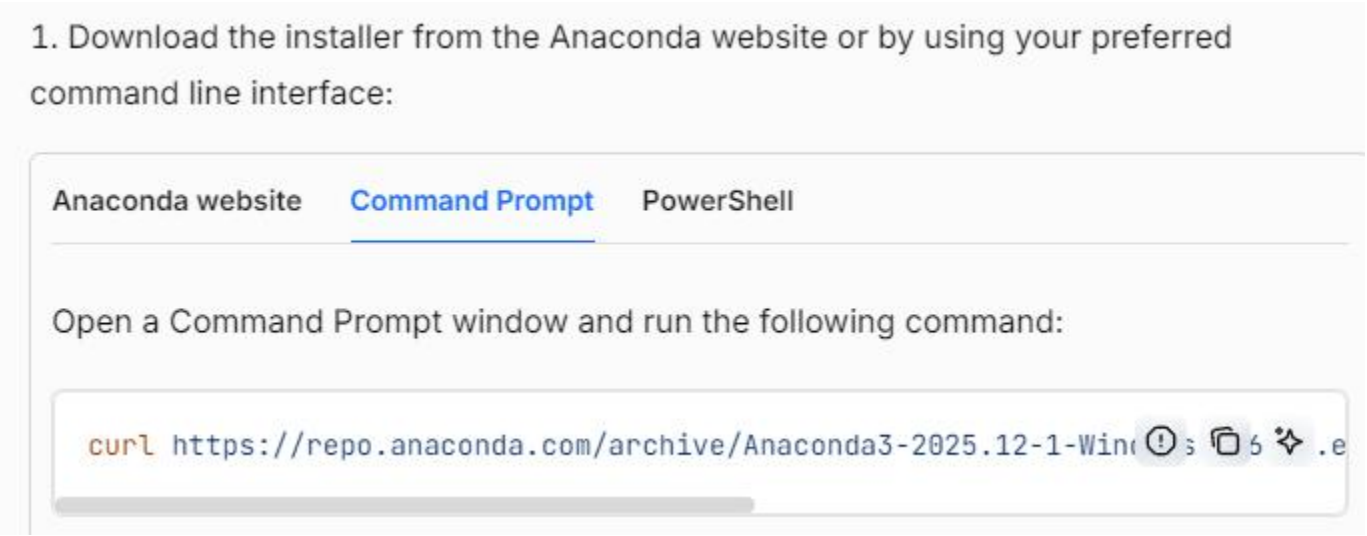
（1）Python environment setup



https://www.anaconda.com/docs/getting-started/main

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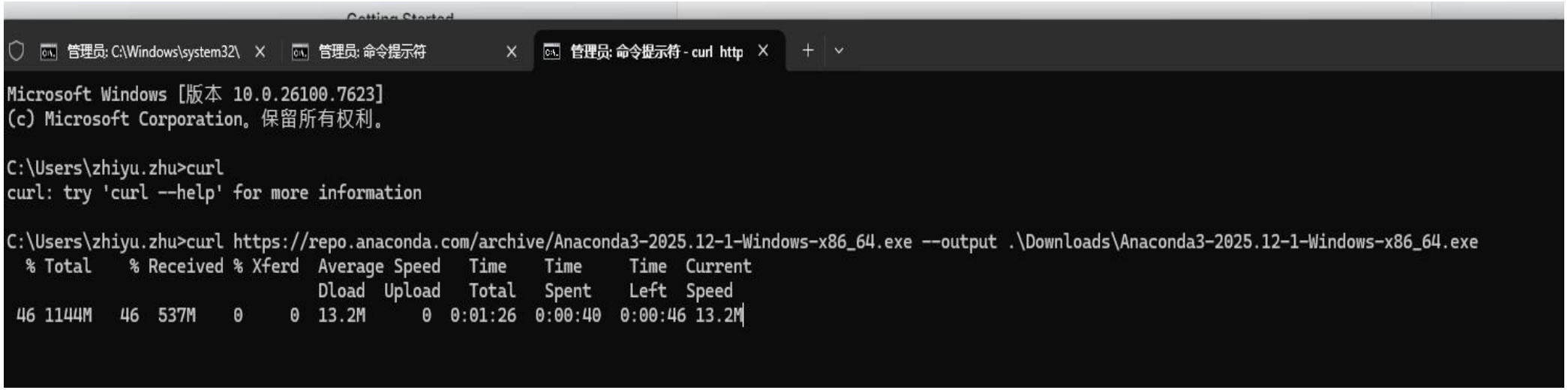
（1）Python environment setup



curl https://repo.anaconda.com/archive/Anaconda3-2025.12-1-Windows-x86_64.exe --output .\Downloads\Anaconda3-2025.12-1-Windows-x86_64.exe

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（1）Python environment setup

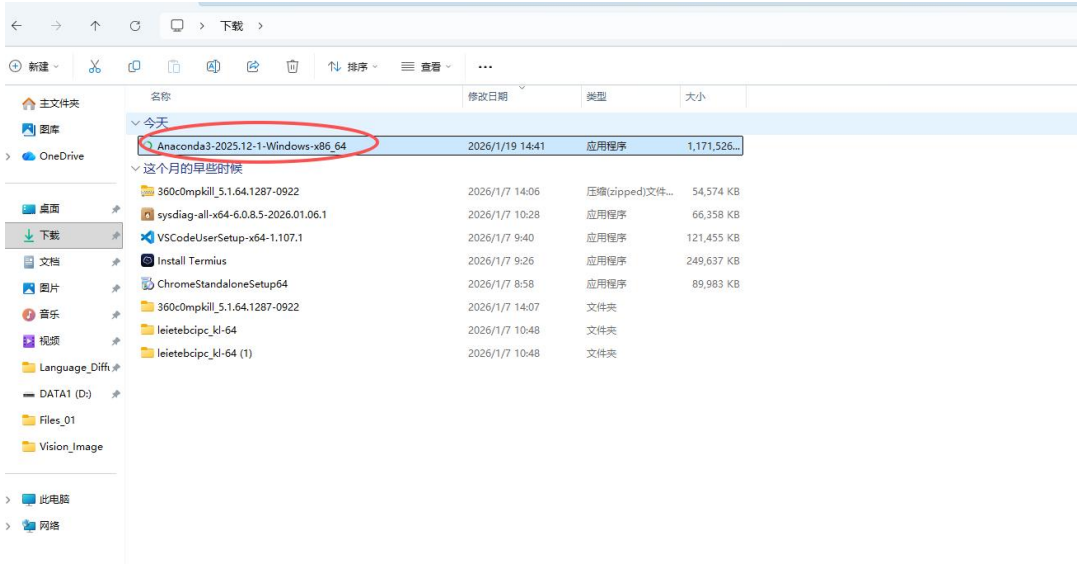


（1）Python environment setup

3. Go to your Downloads folder (or Home folder if downloaded via CLI) and double-click the installer to launch.

> ⚠ To prevent permission errors, do not launch the installer from the **Favorites folder**.

> ⓘ If you encounter issues during installation, temporarily disable your anti-virus software during install, then re-enable it after the installation concludes. If you installed for **All Users**, uninstall Anaconda and re-install it for **Just Me** only.



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（1）Python environment setup

4. Click **Next** and then **I Agree** to agree to Anaconda's Terms of Service (TOS). You can read the full TOS at https://anaconda.com/legal.

5. Select an installation option:

- **Just Me** (Recommended): Install Anaconda Distribution for the current user account.
- **All Users**: Install Anaconda Distribution for all user accounts on the computer (requires Windows Administrator privileges).

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（1）Python environment setup

6. Click **Next**.

7. Select a destination folder to install Anaconda, then click **Next**.

⚠
- Anaconda recommends installing Anaconda Distribution in a directory with no spaces or special characters to avoid potential compatibility issues with open-source tools. For more information, see the **FAQ**.
- Do not install as Administrator unless admin privileges are required.

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（1）Python environment setup

8. Customize your installation options:

- Create shortcuts - Selected by default. Creates Start Menu shortcuts for the Anaconda Navigator, Spyder, Jupyter Notebook, and Anaconda Prompt packages. Deselecting this option skips creating these shortcuts.
- Add Anaconda3 to my PATH environment variable - Adds the path that contains the conda binaries to your PATH environment variable. Anaconda **does not** recommend selecting this option. The conda binaries path contains other package binaries, which are permanently added to your PATH environment variable, even if no conda environment is currently active. This makes it possible for other software to use these package files, which might lead to errors.

  > Unless you plan on installing and running multiple versions of Anaconda Distribution or Python, open Anaconda Navigator or the Anaconda Prompt from the Start Menu to begin your environment management work.

  > As of `Anaconda Distribution 2022.05`, the option to add Anaconda to the PATH environment variable during an **All Users** installation has been disabled. This was done to address **a security exploit**. You can still add Anaconda to the PATH environment variable during a **Just Me** installation.

- Register Anaconda3 as my default Python 3.12 - Selected by default. Registers the Python package in this install as the default Python for programs like VSCode, PyCharm, and so on.
- Clear the package cache upon completion - Runs `conda clean --all --force-pkgs-dirs` after the install finishes. For more information on these commands, see the conda command documentation.

9. Click **Install**. The installation might take a few minutes to complete. Click **Show details** to view the packages being installed.

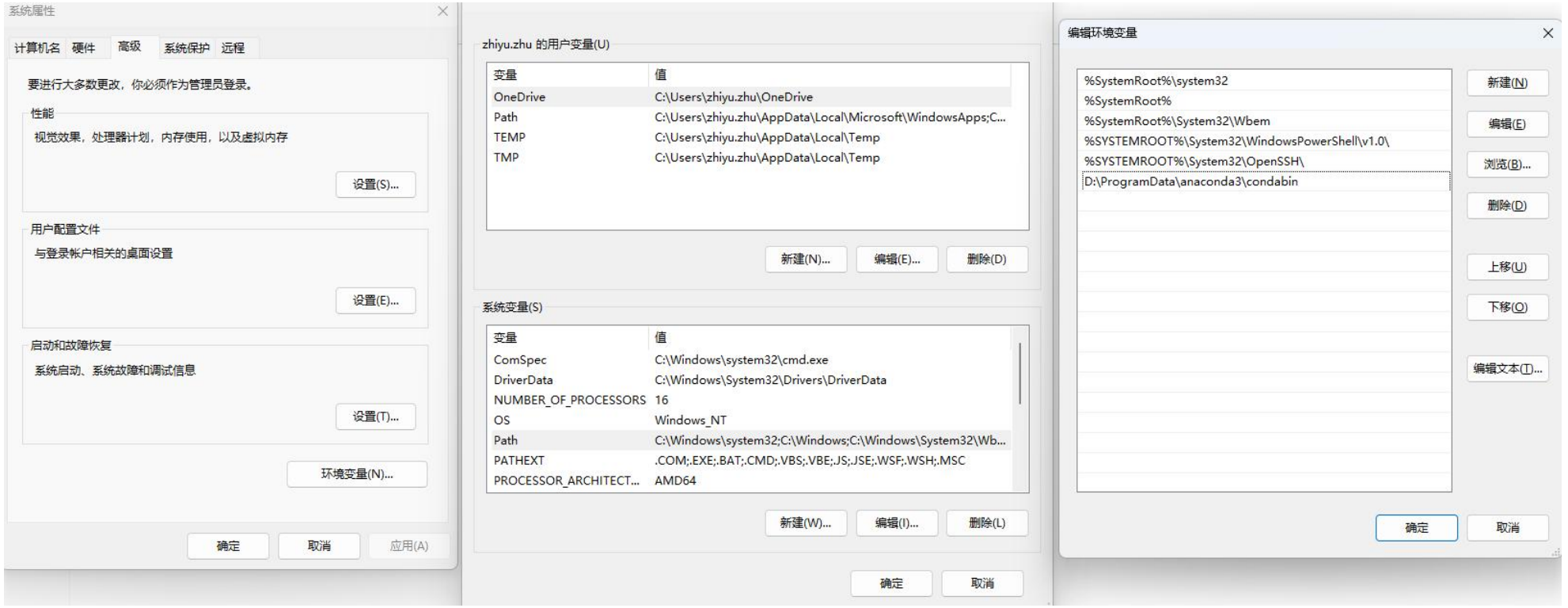
10. Click **Next** twice, then click **Finish** to close the installer.

11. Open Anaconda Prompt to use Anaconda Distribution.

> If you are behind a company proxy, you might need to do some additional setup. For more information, see Using Anaconda behind a company proxy.

（1）Python environment setup

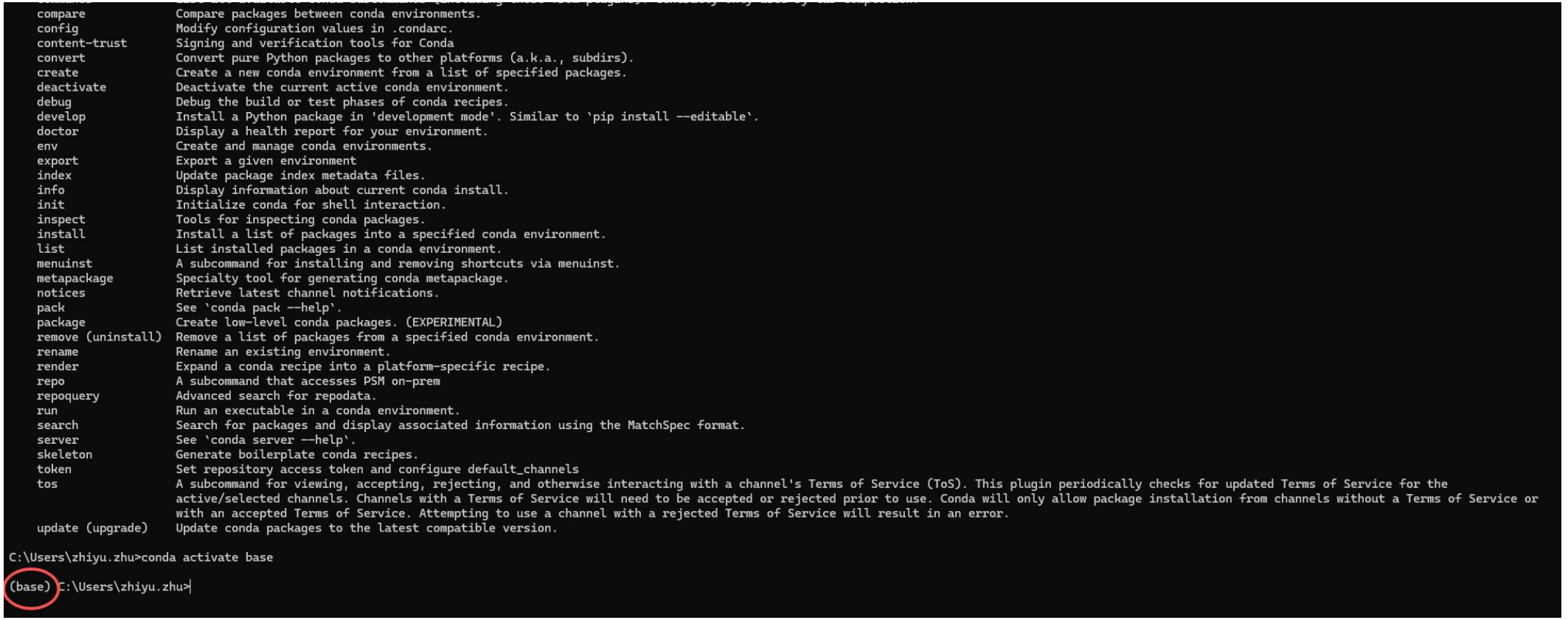
Add anaconda to system variable.



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（1）Python environment setup

Activate base environment or set up your own environment



（1）Python environment setup

```
(base) C:\Users\zhiyu.zhu>conda create -n my_py python=3.12
3 channel Terms of Service accepted
Retrieving notices: \
```

```
The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    expat-2.7.3                |       h885b0b7_4          31 KB
    libexpat-2.7.3             |       h885b0b7_4         119 KB
    python-3.12.12             |       h716150d_1        15.1 MB
    setuptools-80.9.0          |  py312haa95532_0         1.8 MB
    sqlite-3.51.1              |       hda9a48d_0         917 KB
    vc-14.3                    |      h2df5915_10          19 KB
    wheel-0.45.1               |  py312haa95532_0         177 KB
    ------------------------------------------------------------
                                           Total:        18.2 MB

The following NEW packages will be INSTALLED:

  bzip2              pkgs/main/win-64::bzip2-1.0.8-h2bbff1b_6
  ca-certificates    pkgs/main/win-64::ca-certificates-2025.12.2-haa95532_0
  expat              pkgs/main/win-64::expat-2.7.3-h885b0b7_4
  libexpat           pkgs/main/win-64::libexpat-2.7.3-h885b0b7_4
  libffi             pkgs/main/win-64::libffi-3.4.4-hd77b12b_1
  libzlib            pkgs/main/win-64::libzlib-1.3.1-h02ab6af_0
  openssl            pkgs/main/win-64::openssl-3.0.18-h543e019_0
  pip                pkgs/main/noarch::pip-25.3-pyhc872135_0
  python             pkgs/main/win-64::python-3.12.12-h716150d_1
  setuptools         pkgs/main/win-64::setuptools-80.9.0-py312haa95532_0
  sqlite             pkgs/main/win-64::sqlite-3.51.1-hda9a48d_0
  tk                 pkgs/main/win-64::tk-8.6.15-hf199647_0
  tzdata             pkgs/main/noarch::tzdata-2025b-h04d1e81_0
  ucrt               pkgs/main/win-64::ucrt-10.0.22621.0-haa95532_0
  vc                 pkgs/main/win-64::vc-14.3-h2df5915_10
  vc14_runtime       pkgs/main/win-64::vc14_runtime-14.44.35208-h4927774_10
  vs2015_runtime     pkgs/main/win-64::vs2015_runtime-14.44.35208-ha6b5a95_10
  wheel              pkgs/main/win-64::wheel-0.45.1-py312haa95532_0
  xz                 pkgs/main/win-64::xz-5.6.4-h4754444_1
  zlib               pkgs/main/win-64::zlib-1.3.1-h02ab6af_0


Proceed ([y]/n)? y
```

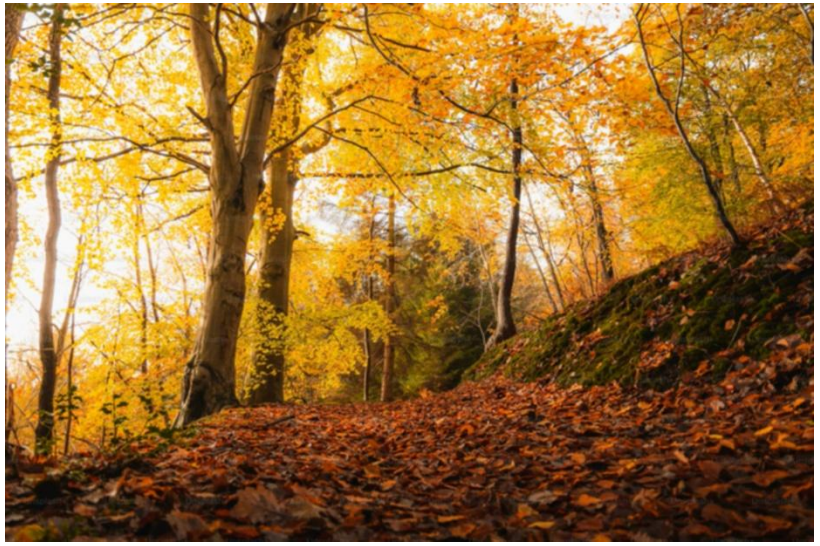
（1）Python environment setup

Installing necessary package. numpy and opencv-python

```
(my_py) C:\Users\zhiyu.zhu>pip install numpy opencv-python
Collecting numpy
  Downloading numpy-2.4.1-cp312-cp312-win_amd64.whl.metadata (6.6 kB)
Collecting opencv-python
  Downloading opencv_python-4.13.0.90-cp37-abi3-win_amd64.whl.metadata (20 kB)
Downloading numpy-2.4.1-cp312-cp312-win_amd64.whl (12.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.3/12.3 MB 19.3 MB/s  0:00:00
Downloading opencv_python-4.13.0.90-cp37-abi3-win_amd64.whl (40.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 40.2/40.2 MB 18.7 MB/s  0:00:02
Installing collected packages: numpy, opencv-python
Successfully installed numpy-2.4.1 opencv-python-4.13.0.90
```

（2）Experiment 1 (average blurring the example image)





```python
import cv2
import numpy as np

# Load the image
image = cv2.imread('input_image.png')

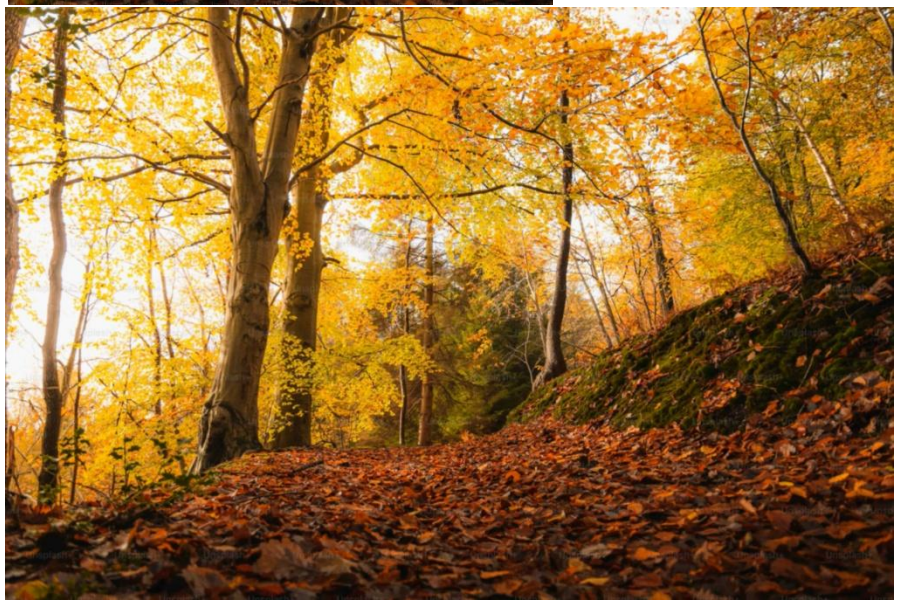
if image is None:
    print("Error: Image not found.")
else:
    # 1. Averaging (Box Filter) Kernel
    kernel_averaging = np.ones((5, 5), np.float32) / 25
    res_averaging = cv2.filter2D(image, -1, kernel_averaging)


    # Save results
    cv2.imwrite('manual_averaging.png', res_averaging)

    print("Manual filtering complete. Images saved.")
```

（3）Experiment 2 (gaussian blurring the example image)

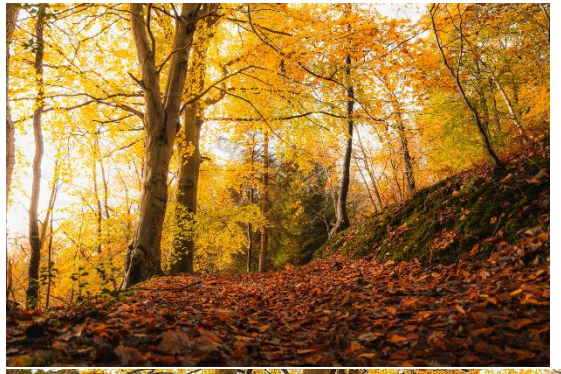


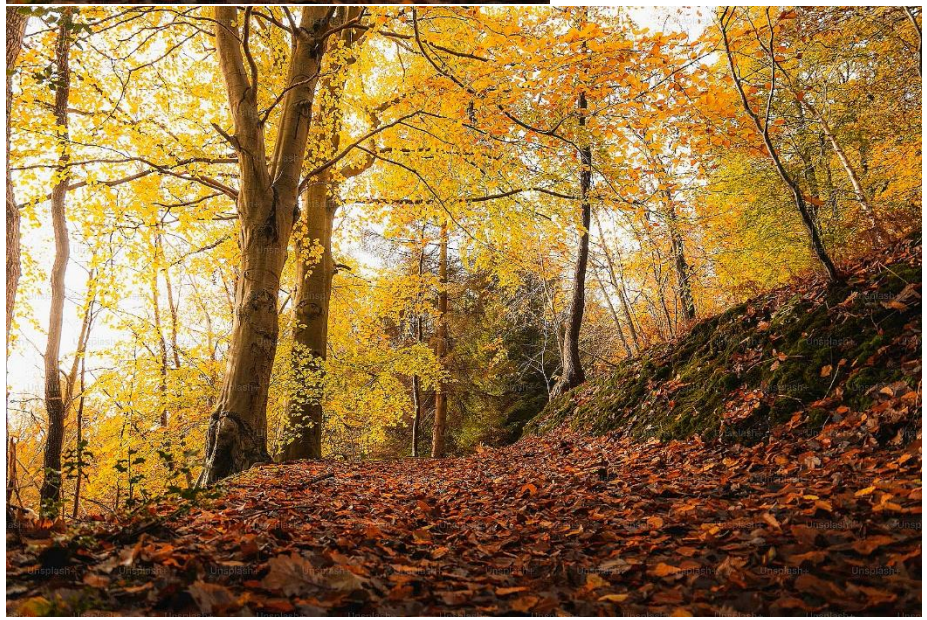


```
# 2. Gaussian Kernel (5x5)
kernel_gaussian = np.array([[1,  4,  7,  4, 1],
                            [4, 16, 26, 16, 4],
                            [7, 26, 41, 26, 7],
                            [4, 16, 26, 16, 4],
                            [1,  4,  7,  4, 1]]) / 273
res_gaussian = cv2.filter2D(image, -1, kernel_gaussian)

cv2.imwrite('manual_gaussian.png', res_gaussian)
```

（4）Experiment 3 (Sharpening for example image)

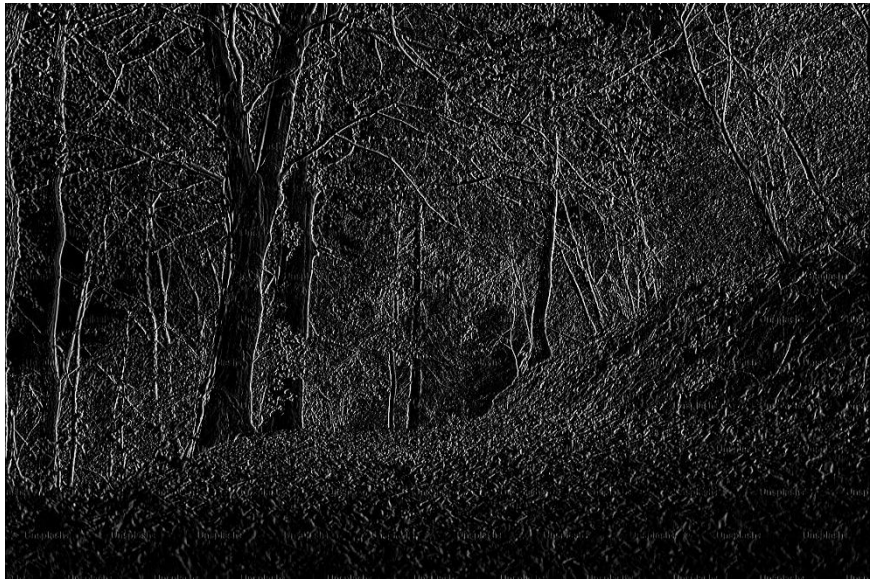




```python
# 3. Sharpening Kernel
kernel_sharpening = np.array([[ 0, -1,  0],
                              [-1,  5, -1],
                              [ 0, -1,  0]])
res_sharpening = cv2.filter2D(image, -1, kernel_sharpening)
cv2.imwrite('manual_sharpened.png', res_sharpening)
```

（4）Experiment 3 (edge filtering for example image





```python
# 4. Edge Detection (Sobel Filter - Horizontal)
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
kernel_sobel_x = np.array([[-1, 0, 1],
                           [-2, 0, 2],
                           [-1, 0, 1]])
res_edges = cv2.filter2D(gray_image, -1, kernel_sobel_x)

cv2.imwrite('manual_edges.png', res_edges)
```

（4）Experiment 4 (Calculating PSNR)

```python
import cv2
import numpy as np
import math

def calculate_psnr(img1, img2):
    if img1.shape != img2.shape:
        raise ValueError("Input images must have the same dimensions.")

    mse = np.mean((img1.astype(np.float64) - img2.astype(np.float64)) ** 2)

    if mse == 0:
        return float('inf')

    max_pixel = 255.0
    psnr = 20 * math.log10(max_pixel / math.sqrt(mse))

    return psnr

original = cv2.imread('input_image.png')
processed = cv2.imread('manual_gaussian.png')

if original is not None and processed is not None:
    score = calculate_psnr(original, processed)
    print(f"PSNR value: {score:.2f} dB")
```

Questions?

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